

FitBuddy – AI Fitness Plan Generator

2. Requirement Analysis Phase

Project Requirements

FitBuddy requires a web interface for collecting user details, a backend for processing requests, Gemini integration for AI-generated responses, and a database for storing users and plans.

Functional Requirements

The system should collect username/user ID, age, weight, fitness goal, and workout intensity; generate a 7-day workout plan; generate a nutrition or recovery tip; accept feedback; update the workout plan; store original and updated plans; and provide an administrator view.

User Requirements

Users should be able to enter their details easily, receive a personalized result, view the workout plan and nutrition tip, and submit feedback for plan refinement.

Admin Requirements

The administrator view should display registered users and their respective original and updated plans.

Technical Requirements

Python, FastAPI, Google Gemini models, HTML/CSS, Jinja2, SQLAlchemy, SQLite, Uvicorn, and Git.

Data Requirements

The application stores user information including ID, name, age, weight, goal, and intensity, along with generated plans and feedback-based updates.

Output Requirements

The output should contain a structured day-wise workout plan and a concise nutrition/recovery tip, with updated results after feedback.

Project: FitBuddy – AI Fitness Plan Generator using Gemini Models.